Smart City Energy Consumption Monitoring and Prediction System: The Case of London

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Abstract

Internet of Things enabled smart meters allow cities to move from reactive energy operations toward proactive, data driven planning. Using the London Smart Meters dataset, this project designs a conceptual Smart City Energy Consumption Monitoring and Prediction System that converts raw meter telemetry into actionable demand intelligence. Key processing steps were conducted on the data, such as normalization, cleaning, augmentation with external data sources and feature engineering. Exploratory analysis showed that demand is driven primarily by seasonality, strong inverse temperature effects, and consistent socio demographic differences across ACORN groups. Building on these patterns, the system integrates two complementary deep learning models: a feedforward neural network that classifies daily demand risk states and an LSTM model that forecasts three day ahead baseline consumption. Results and model outputs are operationalized through an interactive Tableau dashboard that supports monitoring, forecasting, and segment aware planning.

Keywords: IoT, Smart City, Energy Consumption, LSTM, Neural Network, Smart Meters

Smart City Energy Consumption Monitoring and Prediction System: The Case of London

Smart cities increasingly depend on IoT sensing to monitor urban infrastructure at fine temporal resolution, and residential energy demand is a high impact domain for predictive decision making. Smart meters act as distributed sensors that continuously capture electricity usage, enabling cities to understand how demand changes across time, weather conditions, and community segments. However, translating this data into reliable intelligence requires careful handling of dataset structure and bias.

In the London Smart Meters dataset, early aggregate trends can be misleading because total consumption grows as more households enter the program. To ensure analytical validity, this project uses a normalized per household consumption metric as the foundation for all downstream analysis and modeling. Within this normalized framework, exploratory findings show clear seasonal demand cycles, a strong inverse relationship between temperature and consumption, and meaningful differences across ACORN socio demographic groups, indicating that demand reflects the interaction of climate, behavior, and context rather than simple time based growth.

The project objective is to demonstrate an end-to-end Smart City energy analytics workflow that combines IoT system design, behavioral analysis, and predictive modeling. We implement a dual model approach to support operational use cases. A neural network provides a demand risk classification signal for monitoring and alerting, and an LSTM produces a short term three-day-ahead forecast to support planning. These outputs, along with key behavioral indicators, are integrated into a Tableau dashboard that represents the analytics layer of the

proposed system and illustrates how smart meter telemetry can support adaptive urban energy management.

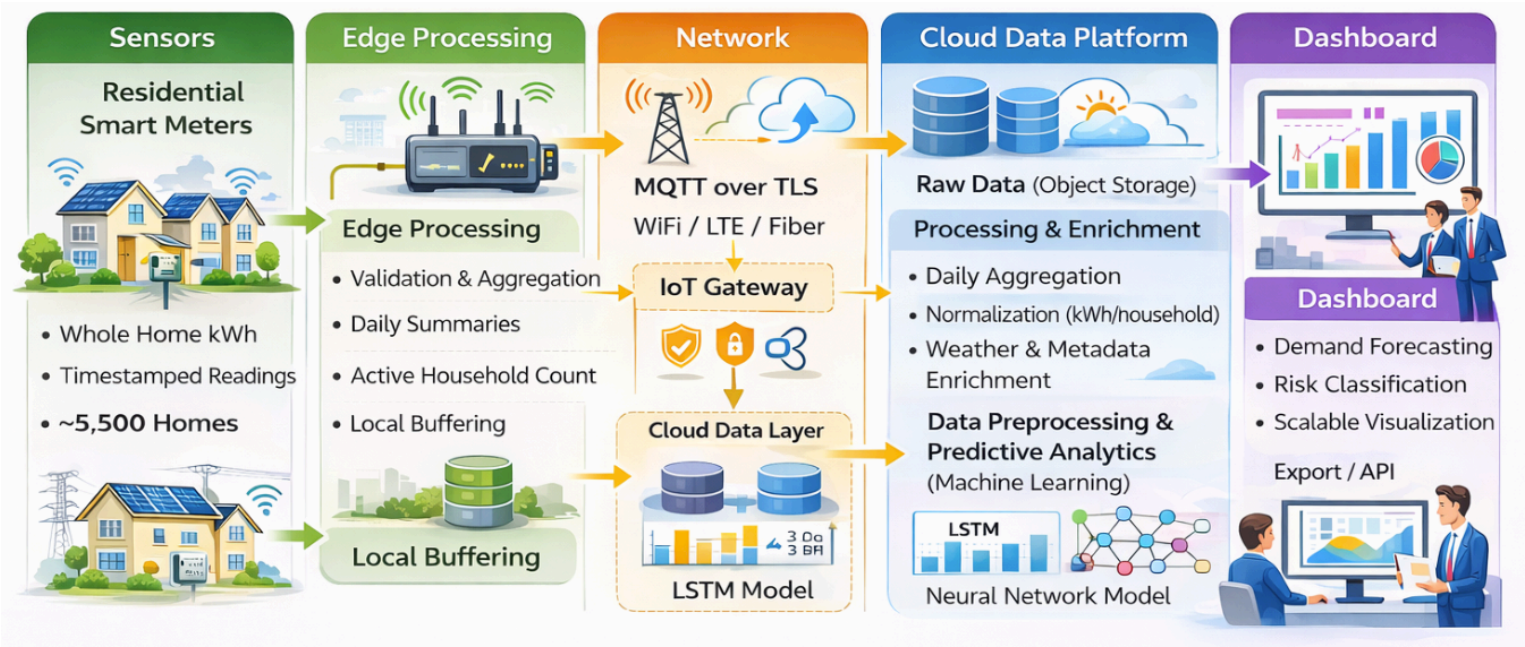
IoT System Architecture

A city scale Smart City Energy Consumption Monitoring and Prediction System is proposed to align with the dataset structure and end to end workflow implemented in this project (Figure 1). The sensing layer uses residential smart meters installed at each home's electrical service entry point to capture timestamped whole home electricity usage, later aggregated to daily kWh for modeling and reporting. Because the signal is whole home only and the active meter population can vary over time, the pipeline supports per household normalization to prevent biased demand trends and is designed to tolerate occasional data loss from interference, variable neighborhood connectivity, and planned or unplanned outages.

At the edge, a neighborhood gateway or utility concentrator aggregates meter telemetry, applies lightweight validation, and provides secure device authentication with short term local buffering to backfill missed transmissions. The gateway generates daily summaries, including an active household count used for normalization and feature generation. Data is forwarded to the cloud over LTE, fiber, or municipal broadband using MQTT over TLS for secure, scalable publish subscribe messaging, with HTTPS reserved for maintenance and backfill transfers when streaming is unavailable. In the cloud data platform, an IoT Hub style ingestion service routes telemetry into streaming and batch workflows, archives raw messages in low cost object storage for traceability and replay, and maintains curated daily tables in an analytics optimized database to support aggregation, normalization, and enrichment with contextual inputs such as daily weather and household segmentation. Predictive analytics are produced in the cloud via two

complementary models, a neural network for demand risk classification and an LSTM for short horizon forecasting, and outputs are delivered to the Tableau decision layer for monitoring and planning (Figure 1). Scalability is achieved through geography or feeder based partitioning and autoscaled processing, enabling expansion from pilot deployments to citywide operation.

Figure 1. *IoT Diagram*

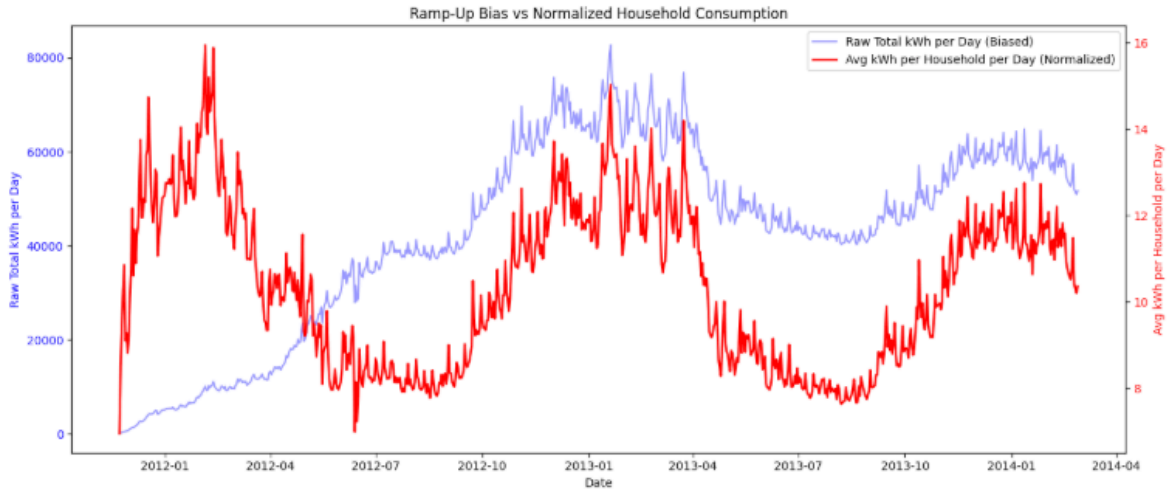


Exploratory Data Analysis: Discovering Behavioral Patterns

Early data understanding showed that raw daily totals misleadingly suggested long term growth, but this trend was primarily caused by the dataset ramp up as more households entered the program. To remove this structural bias, consumption was normalized per household using **avg_kwh_per_household_per_day,** which isolates behavioral demand and keeps values within an interpretable daily range for analysis and modeling (Figure 2). Following normalization, the exploratory analysis showed that residential electricity demand follows a consistent behavioral structure shaped by seasonality, weather, and socio demographic context rather than random fluctuation. Average daily consumption per household peaked in winter, reaching just over about

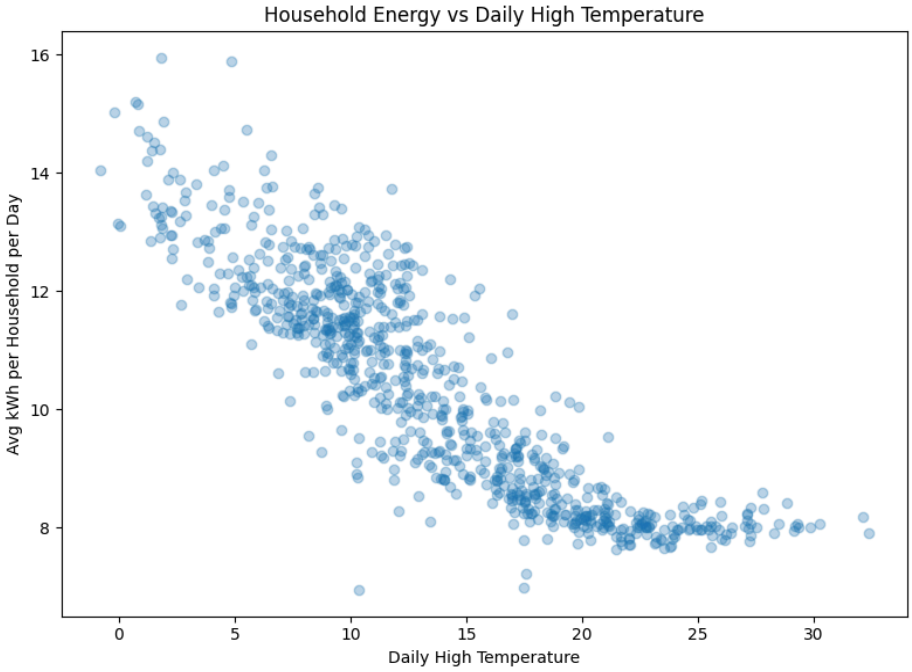
12 kWh per day in January and February, and declined in summer to around 8 kWh per day, consistent with heating driven demand. Weekly effects were present but smaller, with weekends slightly higher than weekdays, suggesting increased occupancy and activity.

Figure 2. *Ramp Up Bias vs Normalized Household Consumption*



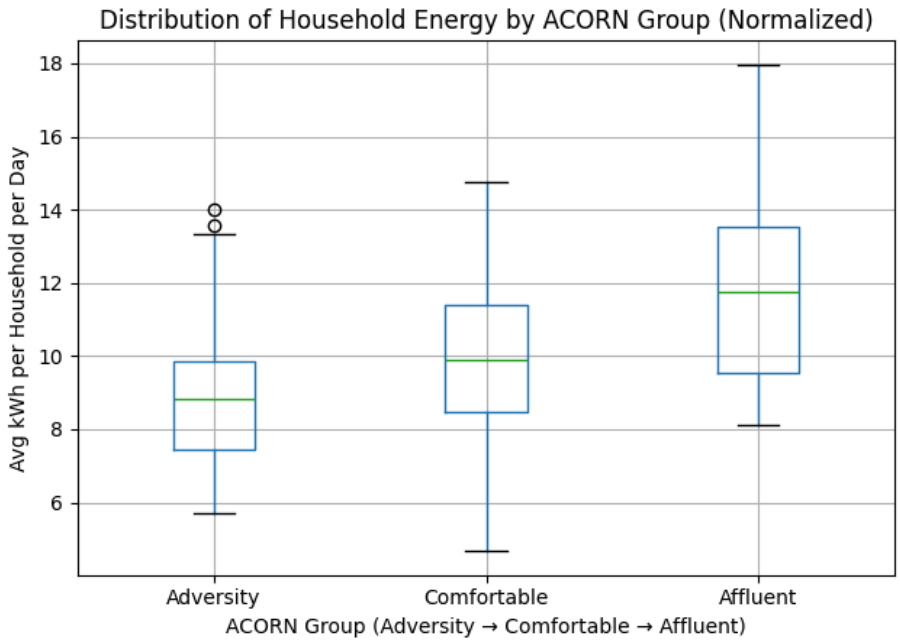
Normalized consumption was then joined with daily weather observations, producing 828 matched daily records with minimal missingness. The results showed a clear inverse temperature relationship, where colder days consistently produced higher consumption (Figure 3). Correlations confirmed temperature as the dominant environmental predictor, with temperatureHigh around negative 0.88 (and similarly strong negative values for other temperature based measures), while humidity and cloud cover showed more moderate associations and pressure and visibility contributed little. These findings support using weather forecasts as an input to Smart City energy monitoring systems to improve demand anticipation and operational planning.

Figure 3. *Temperature vs Consumption Scatter Plot*



Socio demographic segmentation using the ACORN system (A Classification of Residential Neighbourhoods) added a second behavioral layer. After filtering non informative categories, three primary groups remained, Adversity, Comfortable, and Affluent. Average normalized consumption increased monotonically across groups, with Adversity at 8.88 kWh per day, Comfortable at 10.01 kWh, and Affluent at 11.75 kWh (Figure 4). Distributions also showed that affluent households tended to have both higher central consumption and greater variability, consistent with differences in dwelling characteristics and appliance intensity. Interaction patterns reinforced this structure, as all groups increased consumption under colder temperatures, but the magnitude of response differed, with affluent households showing the strongest sensitivity and reaching the highest demand levels under extreme cold conditions.

Figure 4. *ACORN Consumption Comparison Boxplot*



Predictive Modeling

Modeling Approach within the Smart City Framework

The predictive modeling component represents the operational layer of the proposed Smart City Energy Consumption Monitoring System. Rather than developing models solely for analytical experimentation, the modeling strategy was designed to translate behavioral insights from exploratory analysis into actionable intelligence that supports monitoring, forecasting, and planning within an IoT energy management context. Earlier analysis demonstrated that normalized household electricity consumption is primarily influenced by seasonal temperature variation, socio demographic segmentation, and temporal behavior patterns. Based on these findings, two complementary deep learning models were implemented to address distinct operational needs within the IoT architecture, a neural network for demand risk evaluation and a sequential LSTM model for short term energy forecasting. Together, these models provide both

real time situational awareness and forward planning capability aligned with smart grid decision workflows.

Neural Network Model: Demand Risk Evaluation

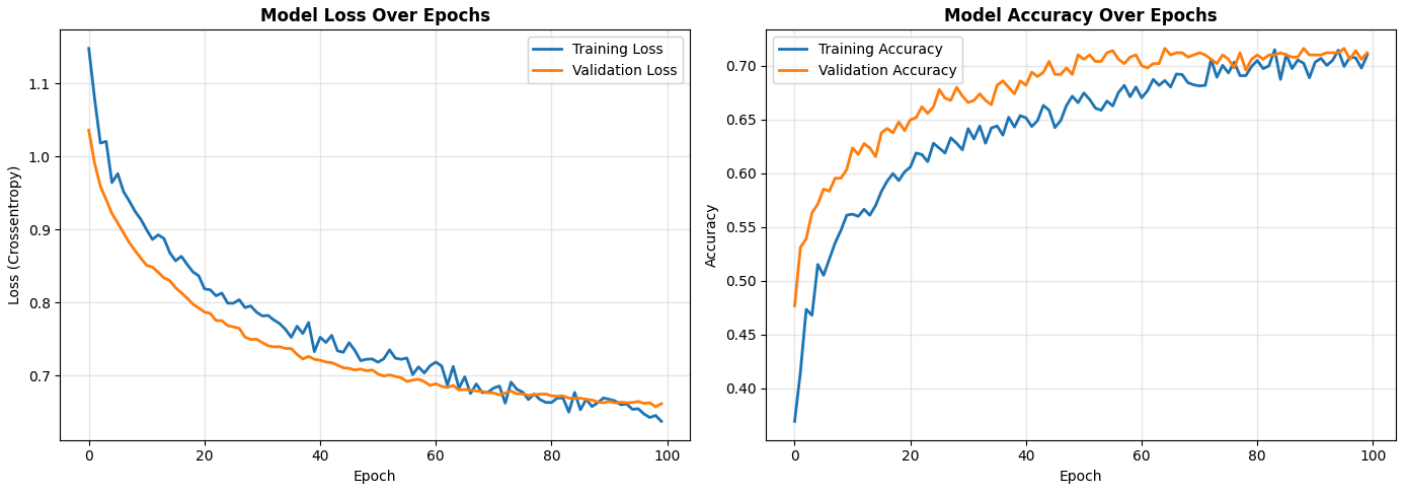
The neural network model was developed to identify elevated demand conditions that may indicate potential peak risk periods. In the context of smart city energy monitoring, classification of demand risk is often more actionable than predicting exact consumption values because it enables operators to recognize emerging system stress conditions early. The target variable was defined by calculating the percentile rank of daily energy usage within each unique combination of season and ACORN group, and encoding it into seasonally adjusted Low, Normal, and High risk class labels. Model inputs were derived from 12 key behavioral drivers identified during analysis, including seasonal temporal indicators, weather variables (such as high and low temperatures, humidity, and wind speed), and encoded ACORN socio demographic categories. This feature design ensures that the model aligns directly with the project objective of integrating environmental, social, and policy variables into predictive decision making.

A feedforward neural network was implemented in TensorFlow using chronological data splitting to preserve temporal order. The architecture included dense nonlinear layers of 64 and 32 units, both utilizing ReLU activation and dropout regularization rates of 0.3 to prevent overfitting. The final output layer utilized a three unit softmax activation to produce multi class probabilities for the three consumption tiers.

Training leveraged the Adam optimizer with a learning rate of 0.001 and sparse categorical crossentropy loss over a maximum of 150 epochs. Visual analysis of the training history demonstrated stable convergence, with validation loss stabilizing around 0.66 and

accuracy rising to approximately 0.71, indicating that the model successfully learned generalized behavioral relationships without suffering from severe overfitting (Figure 5). This stability was further supported by automated Early Stopping, which halted training at epoch 130 and restored optimal weights from epoch 115.

Figure 5. *Model 1 – Neural Network Training History*



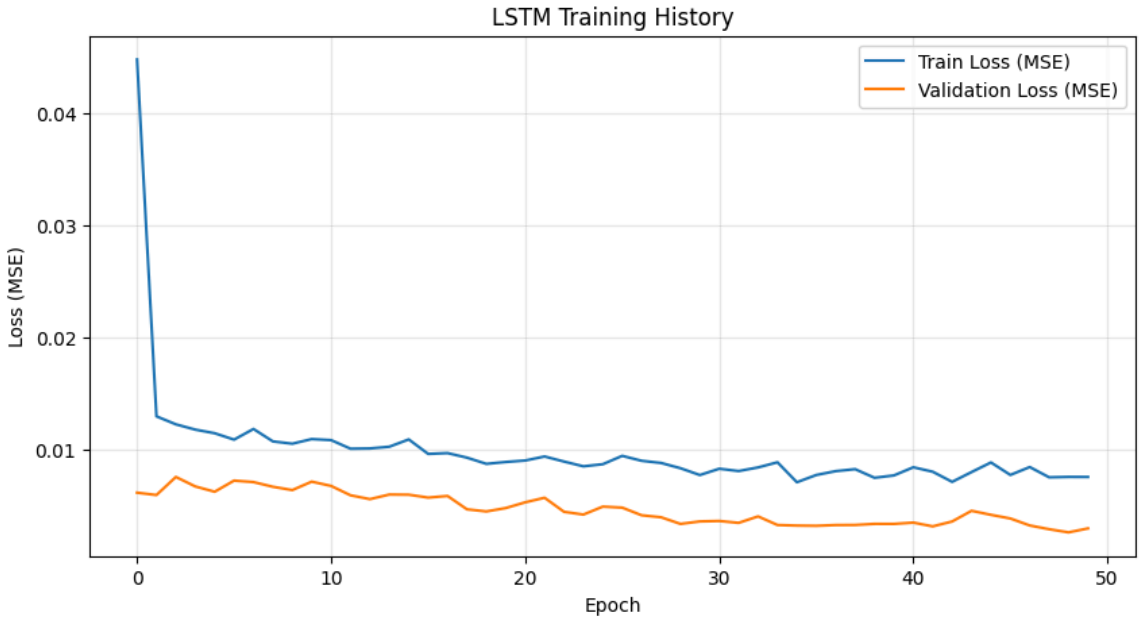
Upon final evaluation on the unseen test set, the neural network achieved an overall accuracy of 0.71, significantly outperforming the random baseline of 0.33. An analysis of the confusion matrix and classification report revealed that the model is highly effective at identifying extreme consumption days, achieving a precision of 0.82 for High risk days and 0.78 for Low risk days. The model also captured High risk days with a recall of 0.73. While performance dropped for the Normal class (precision 0.53 and recall 0.61) due to natural ambiguity and overlap near the tier boundaries, the network proved reliable at its primary task of flagging extremes. Within the IoT system architecture, this model functions as a monitoring layer that continuously evaluates incoming energy data streams and outputs a traffic light style indicator to signal potential high demand conditions, supporting proactive demand management.

LSTM Model: Sequential Energy Consumption Forecasting

To complement the categorical demand risk classification, a Long Short Term Memory (LSTM) model was developed to forecast continuous, future baseline consumption levels. While the neural network identifies risk states, the LSTM quantifies expected baseload energy demand three days into the future (t + 3), enabling highly specific short term grid capacity planning. The forecasting model utilized the normalized average daily energy consumption per household metric, scaled via MinMax normalization to bound inputs between 0 and 1. To capture temporal dependencies, the data was structured using a sliding window approach with a 48 day lookback period to predict the t + 3 horizon, followed by a strict chronological train test split (70 percent training, 30 percent testing).

A deep, stacked LSTM architecture was implemented to capture both short term weekly fluctuations and long term seasonal dependencies. The network utilized three consecutive LSTM layers containing 75 units each, with the first two returning full sequences, and applied a 0.2 dropout rate after every layer for robust regularization. A final dense layer produced the single step continuous consumption forecast. The model was trained using the Adam optimizer and Mean Squared Error (MSE) loss.

Figure 6. *Model 2 – LSTM Training History*



The plotted training history showed an excellent learning curve, with both training and validation MSE dropping sharply in the initial epochs and stabilizing closely together near 0.007, demonstrating strong generalization (Figure 6). Training was run for up to 50 epochs, and the final model restored best weights from epoch 46 based on validation performance.

When inverse transformed back to real world energy units (kWh) and evaluated on the test set, the LSTM demonstrated excellent predictive capability, achieving an R^2 score of 0.9085. Furthermore, the model achieved a Root Mean Squared Error (RMSE) of 0.4720 kWh and a Mean Absolute Error (MAE) of 0.3555 kWh.

Figure 7. *LSTM Predicted vs Actual Consumption (t+3)*

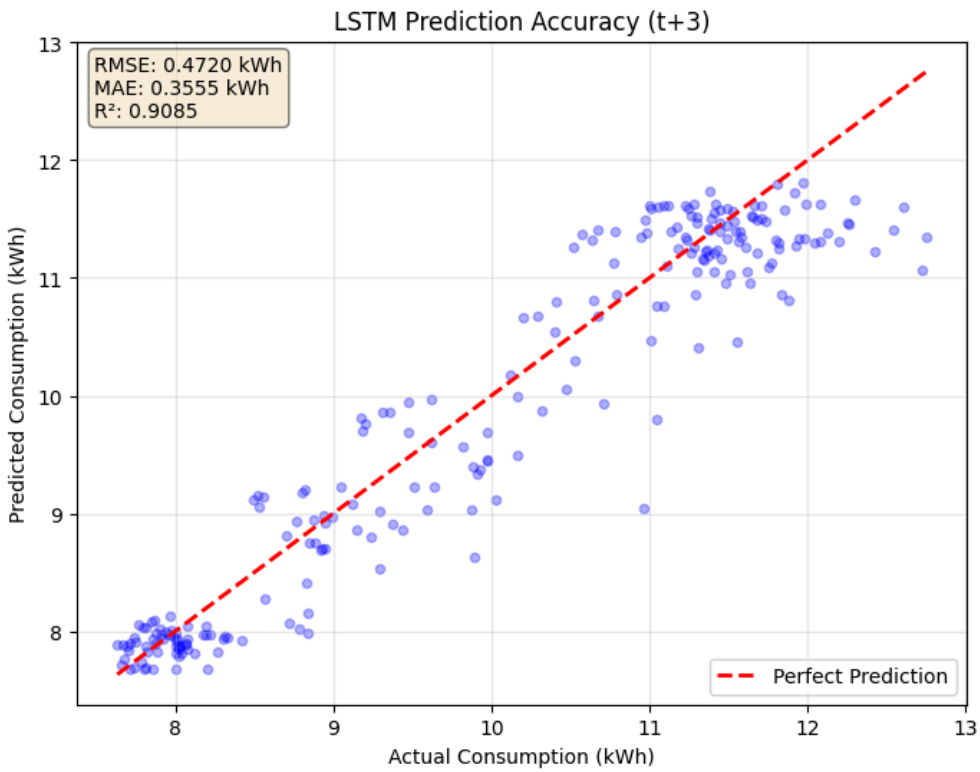


Figure 7 shows the LSTM closely tracking observed consumption across the test period, with the predicted vs actual points clustering near the 1:1 line, indicating strong forecast agreement. With these near term forecasts and demand risk signals generated, the next section

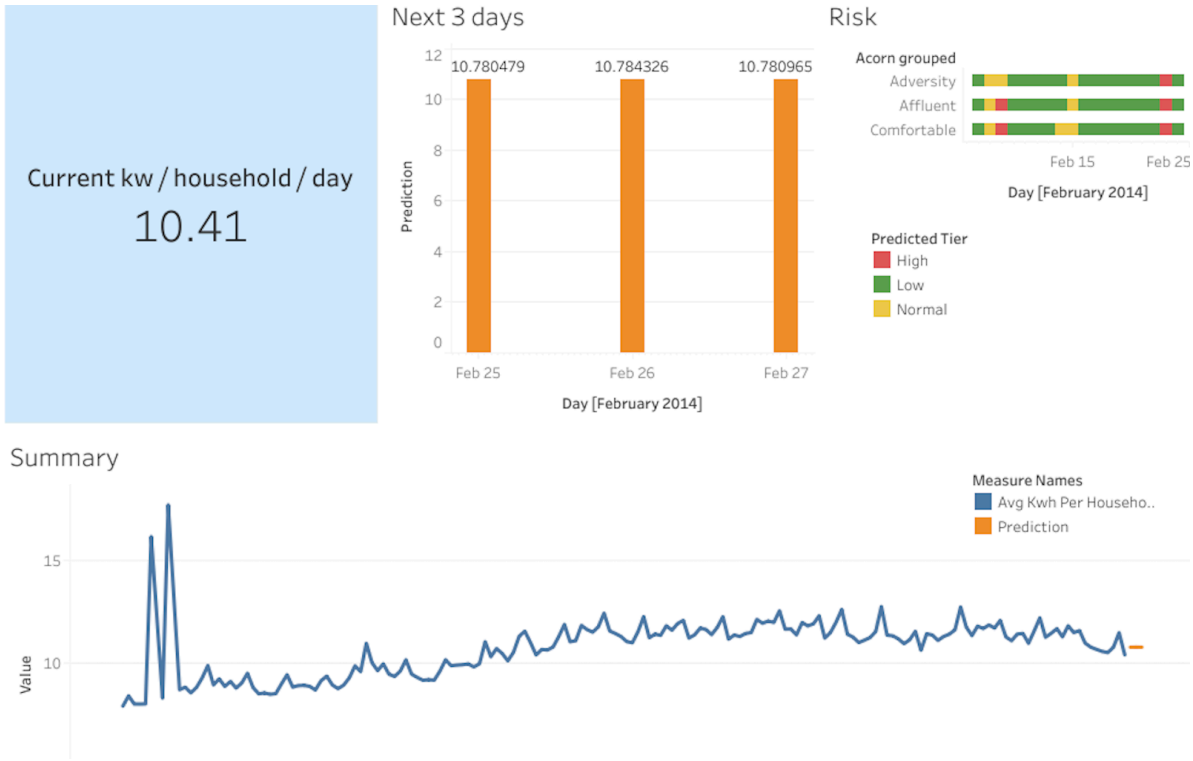
translates both model outputs into an operational Smart City decision layer through an integrated Tableau monitoring dashboard.

Tableau Public Dashboard Visualization

To operationalize the modeling outputs as a Smart City decision layer, an interactive Tableau Public dashboard was built to integrate current conditions, recent history, and near term predictive intelligence in a single monitoring view.

The dashboard includes the following monitoring components. A Current Status KPI reports the latest daily average consumption per household. A Historical Trend view displays the prior 180 days of normalized consumption and overlays the next 3 day LSTM predictions to provide immediate forward visibility. A Forecast Only view isolates the next three predicted days for near term planning. Finally, a Risk Visualization maps the Neural Network's Low, Normal, High demand tier predictions across ACORN groups as a traffic light style signal for elevated demand risk. Together, these views combine categorical peak risk monitoring with continuous short term forecasting to support operational awareness and planning.

Figure 8. *Tableau Dashboard*



Conclusion and Future Research

This project demonstrates how raw IoT smart meter telemetry can be transformed into actionable intelligence for urban energy management. Using the London Smart Meters dataset, we designed a conceptual Smart City Energy Consumption Monitoring and Prediction System. The exploratory analysis revealed that residential demand is driven primarily by seasonal patterns, temperature sensitivity, and socio demographic context rather than simple long term growth. Building on these insights, a dual model deep learning stack was implemented to support operational decision making: a neural network that classifies daily demand risk states for monitoring and alerting, and an LSTM model that forecasts three day ahead baseline consumption for short term planning. Integrated through a Tableau decision layer, the system illustrates how cities can pair descriptive behavioral analytics with predictive tools to support proactive, data driven energy operations.

Future work can extend this framework in several high impact directions. First, forecasting granularity could be improved by moving from citywide averages to localized predictions such as ACORN specific segments, feeders, or substation level load, enabling more targeted planning and load balancing. Second, model performance and robustness could be enhanced by testing alternative architectures and features, including Transformer based time series models, richer weather inputs, and engineered interaction terms that better capture extreme events and shifting seasonal dynamics. Finally, the system could evolve from prediction to intervention by incorporating dynamic pricing and demand response signals such as Time of Use tariffs, EV charging activity, or smart thermostat telemetry, allowing the platform to recommend or automate actions that reduce peak risk and improve grid resilience.

References

Jean-Michel. (n.d.). *Smart meters in London* [Data set]. Kaggle. https://www.kaggle.com/datasets/jeanmidev/smart-meters-in-london

Projects Links

- **Github:** https://github.com/MaximeBoulat/TeamProject-530
- **Tableau Public:** https://public.tableau.com/app/profile/john.bleau/viz/TeamProject_17712350931480/Dashboard1#2

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